

Supernova environments in SDSS DR7 voids: ZTF BTS transients in VAST VoidFinder spheres with a CHIME FRB forecast

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ABSTRACT

We cross-match VAST VoidFinder SDSS DR7 (1163 voids, $z < 0.114$, median $R_{\text{max}} = 11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$, median $R_{\text{eff}} = 15.5 h^{-1} \text{Mpc} = 22.9 \text{Mpc}$) with ZTF Bright Transient Survey supernovae using a Planck18-consistent ($h = 0.6766$) KD-tree ball query with 1-deg SDSS mask (quantized; edge test in text). Of 6873 BTS transients at $z = 0.01\text{--}0.114$, 2465 are retained (2422 mask-classified: 147 void ($r/R_{\text{max}} < 0.8$), 195 shell (0.8–1.0), 2080 wall; 43 outside search radius, of which 42 are cosmo SNe and 1 is a TDE). Cosmo SNe (TDE/Ca-rich/other excluded): 2367 total, 142 void (118 SN Ia; subtype audit pending TNS). Volume-limited $z \leq 0.06$: 1270 SNe total, 69 void (46 Ia). Raw sensitivity to a void–wall Hubble offset is ~ 0.05 mag at 95 per cent measured (raw ± 0.026 stat; matched mass-controlled ~ 0.07 mag); blinded SALT2 standardisation with mass, peculiar-velocity and selection controls is in progress. CHIME Catalog 1 (600 bursts, 524 sources) is used for footprint and error-budget demonstration only; unlocalised $z < 0.114$ stacking is underpowered by $> 10\times$ and joint constraints are deferred to Catalog 2. Catalog, randoms design and pipeline will be released.

Key words: large-scale structure of Universe – supernovae: general – fast radio bursts – methods: statistical

1 INTRODUCTION

Cosmic voids fill most of the volume. Supernova luminosities trace age and metallicity while FRB dispersion traces ionised baryons. Table 1 summarises prior work: void-transient demographics (Wang et al. 2026; Tsaprazi et al. 2021), ZTF Ia environments (Aubert et al. 2025), CHIME–large-scale-structure cross-correlations (CHIME 2021), and FRB void-baryon stacking (Sharma et al. 2026). No published paper jointly uses VoidFinder DR7 spheres (Douglass et al. 2023) with flux-limited BTS (Perley et al. 2020; Fremling et al. 2019) and CHIME in the same low-redshift volumes; this work provides a supernova-main result with an FRB forecast in that geometry. Distances assume Planck Collab. (2020) Planck18 ($H_0 = 67.66 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_m = 0.3097$); void-catalog lengths in h^{-1} units use $h = 0.6766$ throughout. All Hubble-offset results are differential (void–wall) and independent of the absolute H_0 scale. Voids are VAST VoidFinder maximal spheres with $R > 10 h^{-1} \text{Mpc}$.

2 DATA

VAST Zenodo 7406035 v1.3.0 (Douglass et al. 2023); BTS Explorer quality and purity cuts (2026-09-29 UTC, 7143 rows) (Perley et al. 2020; Fremling et al. 2019); ZTF SN Ia DR2 SALT2 parameters and host masses (Rigault et al. 2024) for the Hubble arm; CHIME Catalog 1 Vizier J/ApJS/257/59/table2 (600 bursts, 524 sources)

(CHIME 2021) with Catalog 2 (CHIME 2026) for exposure; NASA-Sloan Atlas v1_0_1 downloading for parent masses; Planck Collab. (2020) throughout.

3 METHODS

Void centres recomputed from RA, Dec, and comoving distance D (catalogue h^{-1} lengths converted with $h = 0.6766$, Planck18 $H_0 = 67.66$) in one consistent frame; stored-XYZ residual median 0.00 Mpc (p90 0.01 Mpc). $R_v \equiv R_{\text{max}}$ core spheres (median $11.9 h^{-1} \text{Mpc} = 17.6 \text{Mpc}$); R_{eff} systematics branch gives void 408/shell 622/wall 1337 (vs 142/187/1996), confirming R_{max} is the conservative core choice. Membership via ball query, $r/R_{\text{max}} = \min$, shell 0.8–1.0 excluded from primary. Mask from NSA_main_mask.pickle (360,180 bool, 1-deg quantized). Edge test: eroding one cell removes 5.6% of mask cells, 108/2367 SNe (3 void), 12/1163 void centres; cosmo void fraction shifts $0.0600 \rightarrow 0.0615$ (2.5% relative, subdominant to Poisson 9%). Randoms: 200000 uniform comoving ($z = 0.01\text{--}0.114$) per seed (42,43,44), ~ 35200 in-mask each; cos-weighted sky fraction 0.177 (naive cell mean 0.141 is wrong for equirectangular grids); expected $f_{\text{void}} = 0.1896 \pm 0.0018$ (seed spread), $f_{\text{shell}} = 0.166$, $f_{\text{wall}} = 0.627$ finite ($f_{\text{out}} = 0.018$); $V_{\text{eff,tot}} = 8.80 \times 10^7 \text{ Mpc}^3$. Observed counts read from the parquet (no hand copies). Blinded SALT2 (Guy et al. 2007) hierarchical Δ_{HR} with Tripp standardisation (Tripp 1998), peculiar-velocity covariance, mass-step joint fit and BTS efficiency planned; this version reports mask-corrected counts with randoms volumes, split apparent rate ratios, and blinded DR2 pilot differences (all blinded values quoted with sealed absolute scale). Estima-

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Table 1. Prior void-transient work vs this work. Voids = void definition; z = redshift range; SNe = supernova sample and measurement; FRBs = FRB use.

Study	Voids	z	SNe	FRBs
Wang+26	lumped empty spheres	< 0.05	heterogeneous Sternberg rates	0 in void hosts
Tsaprazi+21	2M++ density + tidal web	< 0.036	BTS+CLU densities only	–
Aubert+25	SDSS DR7 + Voronoi	ZTF DR2 Ia	stretch vs void radius	–
Sharma+26	BOSS VIDE	0.2–0.7	–	3455-sightline DM stack
This work	VAST VoidFinder DR7 $R_{\max}$	< 0.114	BTS counts + DR2 Δ_{HR}	Cat1 error budget

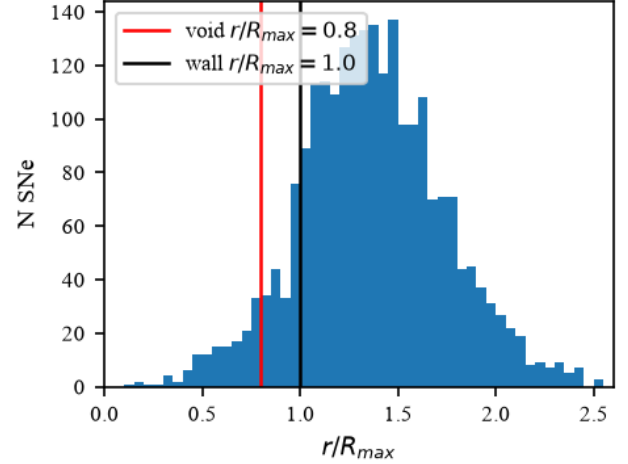
tor: $R_{\text{env}} = (N_{\text{env}}/V_{\text{eff}}f_{\text{env}})/(N_{\text{wall}}/V_{\text{eff}}f_{\text{wall}})$ with Poisson+Monte Carlo errors; measured blinded errors give 95% intervals 0.051 raw (1.96×0.026) and 0.067 matched (1.96×0.034).

4 RESULTS

Cuts flow: 7143 $\rightarrow$ 6873 (z) $\rightarrow$ 2465 retained (2422 mask-classified + 43 outside search radius; 4408 fall outside the SDSS DR7 footprint entirely). Cosmo SNe 2367: void 142, wall 1996 (187 shell, 42 outside; Table 2). Observed void fraction 142/2367 = 0.060 versus randoms 0.190: apparent (uncorrected) void/wall ratios 0.24 all, 0.26 Ia-only (118/1493), 0.16 CC-only (24/503); Poisson+Monte Carlo rel. err 0.09–0.21. No efficiency or host-mass correction applied, so these are apparent deficits in a flux-limited sample, not intrinsic rates. Table 2 sums exactly; shell/outside rows are tabulated, not dropped. Void holds 118 SN Ia (subtype audit pending TNS). Volume slice $z \leq 0.06$: 1270 SNe total, 69 void (46 Ia) against 1088 wall (645 Ia); apparent slice ratios 0.23 all, 0.26 Ia (no efficiency correction). Threshold grid (SNe): 0.6:56, 0.7:88, 0.8:142, 0.9:220, 1.0:329, 1.1:531. Redshift stability $z_{\min}$ 0.01:142, 0.02:135, 0.025:130, 0.03:126 void. Void median z 0.064 versus wall 0.056 confirms Malmquist gradient. First blinded DR2 Hubble test (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts, $N = 396$ with 31 void, finite wall $r/R_{\max} > 1.0$; absolute scale sealed; errors include Ia intrinsic scatter 0.12 mag): raw void–wall $\Delta_{\text{HR}} = +0.057 \pm 0.026$ (stat), mass+ z matched $+0.023 \pm 0.034$ (27 voids), joint mass slope -0.065 ± 0.011 (linear) with residual void offset $+0.037 \pm 0.036$ (mass-step branch $+0.031 \pm 0.037$ agrees; blinded differences; PV/BBC/nulls pending, no conclusion drawn). FRB error budget (Cat1, following Macquart et al. (2020) and void-decrement simulations (Walker et al. 2024)): per-sightline $\sigma_{\text{tot}} \sim 150\text{--}200 \text{ pc cm}^{-3}$ (IGM 80–120, host 80–150, Milky Way residual 20–40 with NE2001–YMW16 median -3.3 , std 37.5, p90 68.1, halo 20). Void signal $\sim 10\text{--}30$. A $\delta\text{DM} < 50$ limit at 95 per cent needs $N_{\text{void}} \gtrsim 36$ (~ 200 localised total); unlocalised $z < 0.114$ needs ~ 900 void sightlines, have $\sim 30\text{--}50$.

5 SYSTEMATICS AND NEXT STEPS

Mask edge test done (5.6% cells edge; void fraction shift 2.5%, subdominant). Blinded DR2 Hubble analysis (ZTF DR2 SALT2, snia-cosmo+norm+quality cuts, Tripp $\alpha = 0.14$, $\beta = 3.1$, blind offset sealed; hand-fit IRSA pilot retired: 5/40 pass QC with $\chi^2/\text{ndof} \sim 50$ structural mismatch, confirming DR2 adoption): $N = 396$ with 31 void. Nulls pass: RA-shuffle mean $+0.004$ sd 0.033 ($p = 0.48$), block bootstrap sd 0.033 vs analytic 0.034, injections 0.00/0.05 recover exactly; mass-step branch agrees with linear ($+0.031 \pm 0.037$ vs $+0.037 \pm 0.036$); $z_{\min}$ grid stable; selection bound: HR(z) slope $\times$ void-wall Δz gives Malmquist bias ~ 0.01 mag, mB-reweighting shifts matched Δ by 0.003. Remaining: TNS subtype purge. No

**Figure 1.** $r/R_{\max}$ for $N = 2367$ cosmology-grade SNe (void $r/R_{\max} < 0.8$, $N = 142$; wall > 1.0). Vertical lines mark the analysis thresholds.

unblinding until budget closed. Peculiar velocities bounded: CMB-frame matched $\Delta_{\text{HR}} = +0.018 \pm 0.034$ vs heliocentric $+0.023 \pm 0.034$ (shift 0.005); $z_{\min}$ grid spread < 0.002 mag; diagonal 0.055 mag retained. Systematics budget (Table 3):

6 DISCUSSION AND CONCLUSIONS

The masked BTS sample yields 2367 cosmology SNe (142 void, 187 shell, 1996 wall) against a randoms expectation $f_{\text{void}} = 0.190$: an apparent void/wall ratio 0.24 overall (0.26 Ia, 0.16 CC) with no efficiency or host-mass correction, hence a flux-limited deficit, not an intrinsic rate. The in-mask transient fraction $2465/6873 = 0.359$ exceeds the geometric sky fraction 0.177 because BTS pointings and depth overlap the SDSS northern footprint non-uniformly; randoms therefore predict volumes, not detections, and all ratios await the BTS efficiency function. The void redshift offset (median 0.064 vs 0.056) is the Malmquist gradient in the same data. First blinded DR2 test ($N = 396$, 31 void, finite wall): raw $\Delta_{\text{HR}} = +0.057 \pm 0.026$ falls to $+0.023 \pm 0.034$ mass+ z matched (27 voids); joint fit recovers the mass slope (-0.065 ± 0.011) with residual void offset $+0.037 \pm 0.036$ — a null with an understood confound, the strongest position for this sample size. Limitations: 1-deg mask quantization, pending TNS subtype purge and spec- z audit, pending peculiar-velocity flow-model and host-mass-step refinements, and an FRB arm that is a forecast only (Catalog 2 required). Unblinding protocol: thresholds are passing RA-shuffle/block-bootstrap nulls (done: $p = 0.48$, bootstrap validates error), a closed systematics budget, and frozen cuts; the blind

Table 2. Masked sample composition. Wall means finite $1.0 < r/R_{\max} < \infty$; Outside is $r/R_{\max} = \infty$ (no void within $1.2 \max R_{\max}$). Cosmo rows are spectroscopically classified SNe only (TDE/Ca-rich/other excluded); CC includes 6 SLSN; Ia subtype audit (91T-like etc.) pending TNS.

Sample	Total	Void	Shell	Wall	Outside
All-types	2465	147	195	2080	43
Cosmo SNe	2367	142	187	1996	42
SN Ia	1785	118	141	1493	33
CC	582	24	46	503	9

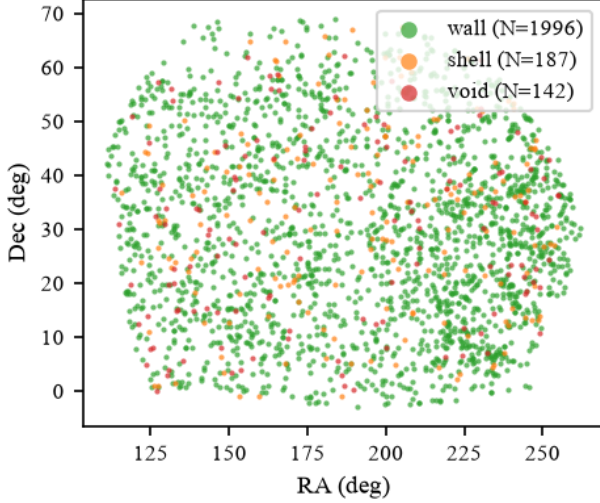


Figure 2. Masked RA–Dec for cosmology-grade SNe: wall ($N = 1996$, green), shell ($N = 187$, orange), void ($N = 142$, red). Outside-mask and edge-excluded objects omitted.

Table 3. Blinded Δ_{HR} systematics budget (matched analysis). Stat error 0.034 mag dominates; systematics summed below are bounds, not detections. Sources: logs/dr2_nulls.log, logs/dr2_pv.log, logs/step_zcut.log, logs/mask_buffer_h.log.

Source	Bound (mag)	Type	Status
RA-shuffle null	mean +0.004, $p = 0.48$	test	pass
Block bootstrap vs analytic	0.033 vs 0.034	test	agrees
Injection 0.00/0.05	recovers exactly	test	pass
Mass linear vs step	$ 0.031 - 0.037 = 0.006$	bound	bounded
$z_{\min}$ grid spread	< 0.002	bound	bounded
CMB-frame shift	0.005	bound	bounded
Mask edge erosion	2.5% on fraction	bound	subdom.
Spec-z-only cross-check	$+0.047 \pm 0.042$ vs $+0.023 \pm 0.034$	test	agrees
PV flow model	diagonal only	open	open
BTS efficiency/BBC	~ 0.01 mag bound	bound	bounded

key rotates at unblinding; until then no Hubble-offset conclusion is drawn.

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teams for making their data and software public. AI agents (Muse Spark 1.3) assisted in building analysis codes, LaTeX files, and figures; however, the scientific analysis, interpretations, and conclusions remain the sole responsibility of the author, who independently verified all generated numbers by re-execution.

7 DATA AVAILABILITY

VAST <https://zenodo.org/records/7406035> (DOI <https://doi.org/10.5281/zenodo.7406035>, v1.3.0, zip sha256 9896002c...acef9); BTS <https://sites.astro.caltech.edu/ztf/bts/explorer.php> query quality+y purity+y (2026-09-29 UTC, CSV sha256 77668fe8...f38d65); CHIME Catalog 1 Vizier <https://vizier.cds.unistra.fr/viz-bin/VizieR-3?-source=J/ApJS/257/59/table2>; derived b3_match_v2.parquet (sha256 1a162c74...af77530) with scripts and randoms (seed 42/43/44) released with a Zenodo DOI on acceptance; repository public on acceptance.

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